

1a. Introduction

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1a. Introduction: outline

- Team
- Location
- Schedule
- Assessments
- Remember

Location

- Lecture
 - Venue: Woolnough Lecture Theatre
 - Time: Mondays 1pm - 3pm
- Labs
 - Lab 1: EZONE CENT 209 (Mondays 3 - 5pm)
 - Lab 2: EZONE NTH 111 (Tuesdays 9-11am)
 - Lab 3: EZONE CENT 209 (Tues 12 - 2pm)
- Consultation
 - MATH 2.21, Thurs 4-5pm

Announcements

- You should sign up to **Moodle** (<https://quiz.jinhong.org>)
 - Then self-enrol in the CITS3006 area
- This will allow you to access the announcements and discussion forum
- We will also use Moodle for submission of assessments
- And, finally, it will be the primary source of information on your teams for the group project

Course overview

Labs start first

Week	Lecture	Lab	Assessments
1	Ethics + Reconnaissance		
2	Network Exploits + Malware	Lab 0: Setup and Linux	
3	More Malware	Lab 1: Network Exploits	
4	Reverse Engineering		Lab Quiz 1
5	Rowhammer (TBC)	Lab 2: Malware	Lab Quiz 1 Demo
6	Privilege Escalation	Lab 3: Reverse Engineering	

Course overview

Week	Lecture	Lab	Assessments
7	Command and Control		Lab Quiz 2
8	Web Security	Lab 4: Privilege Escalation	Project out
9	Active Directory	Lab 5: Web Security	
10	TBC (is a public holiday)		Lab Quiz 3
11	TBC (possibly OSINT)		Lab Quiz 3 Demo
12	Revision		Project demo

Assessments

Assessment Item	When	Covers	Worth (total)
Lab Quizzes	Weeks 4, 7, 10	Materials up to the assessment week (excluding)	60% (20% each)
Group Project	Released week 8	Various	40%

Please note, all dates are tentative and subject to change!

How does the lab quiz work?

- We will be using two formats for the lab quizzes:
 - Complete given tasks over the set amount of time, and do the demo later (the following week lab)
 - Complete given tasks and demo during the scheduled lab
- For the latter method, you will be supervised by a lab facilitator to complete a timed lab quiz (~60 mins but can vary).
- You cannot receive mark if your attendance to the demo is not confirmed by the lab facilitator for both format.
 - i.e., you must be supervised/invigilated by the lab facilitator to receive marks.
 - Attendance is required for all F2F students.

How does the lab quiz work?

- For the demo, we will be using a peer-marking system, where you will be formed into a group of 4 (or thereabout), and each person will have ~15 mins to do the demo.
- Marking keys will be provided for peer markers to use and submit the marking.
- The peer markers will also provide feedback to the demonstrator including the mark range.
- Facilitators will be available for any moderations and review of the peer marking process.
- All submitted marks will be moderated based on the report submitted, and finalised marks will be uploaded to csmarks.

Group Project

- Details to be confirmed
 - Groups will be pre-assigned
 - You will complete several tasks, and submit accompanying reports (group and individual)

Prerequisite

- Prerequisites: CITS2002 or CITS2005
- Recommended: CITS2006 and CITS3002
- This unit requires the student to be knowledgeable in various aspects of CS.
- This unit expects students to be knowledgeable in various programming languages such as Java, Python and C (or be able to quickly catch up as necessary).
- You should also be familiar with basic discrete mathematics and number theory.
- I highly recommend taking this unit in third year after taking various CITS units (including ones listed above) for preparations.

Please note

- This unit is/does NOT:
 - cover a comprehensive penetration testing techniques
 - Pentesting and cybersecurity is a broad discipline!
 - cover all of the latest and greatest attacks
 - Covering selected but important fundamental themes
 - Read online sources instead to keep up to date
 - cover ethical, legal or economic issues
 - We will touch on ethical issues briefly later, but not focus on them

Please note

- Lectures provide **theoretical** and **conceptual** understandings of the topics presented
 - But we will also do some practicals in the lectures
 - So, you should bring your laptop if you want to try them in class as well
- Labs and project provide **practical skills** of the topics presented
- The contents in Lectures, labs and project will be related, but they are all **INDEPENDENT** learning materials
 - i.e., you will be learning new things in each lecture, lab and project.

FAQ

- Why is the lab quiz timed?
 - Being able to solve problems in a reasonable time is a key ability in any roles. We have more flexible assessment – group project that runs over multiple weeks.
 - Also, assessment contents are set such that it is reasonable to solve within the given timeframe.

FAQ

- Some instructions in the assessments seem overly broad, why?
 - This leaves sufficient room for exploration. Also, in any software development/testing environments, you will never have perfect requirements. It is up to you to consult with the client to check that the requirements are met or make reasonable justifications that would be acceptable by the client.
 - The same applies in the assessments - you are also told to discuss with me if any confusions arise, and reasonable and practical justifications and assumptions are accepted for design choices of your solutions.

FAQ

- Why don't we learn about tool X and tool Y?
 - The goal of the unit is not to train you to use specific tools – those are always changing - but rather to focus on the fundamental aspects of the topics covered.
 - 10 years ago, none of those tools existed (or most).
 - 10 years later, you probably be using some other tools.
 - However, the fundamentals don't change quickly.